READ-ONLY APP GUIDE

SatQuery AI

How it works, how to use it, what it produces, and how to present it with confidence.

What this guide is
An operating manual, technical summary, output-reading guide, and presentation brief.

Review scope
Read-only inspection of the repository as it exists on 16 Oct 2026. No application source or configuration was changed.

Prepared for a demo, jury review, and team handover.

1. What SatQuery AI is

SatQuery AI is a local web application for asking plain-language questions about remote-sensing imagery. A user supplies one scene, an optical-plus-SAR pair, or two dates of the same area. The app validates the files, selects a fixed tool plan, combines deterministic image measurements with a vision-language model, and returns an answer with masks, numbers, and an execution trace.

The one-sentence pitch
SatQuery turns a satellite-image question into an evidence-backed answer: not only a sentence, but imagery layers, pixel measurements, tool calls, and an exportable audit trail.

What is in the current repository

Area	Implemented now	Safe way to describe it
Core analysis	Single-image VQA, captioning, grounding, bi-temporal change analysis, and optical-plus-SAR fusion.	A working prototype with deterministic RS tools plus a zero-shot VLM.
Evidence	Red mask overlays, measured statistics, input metadata, confidence, and an expandable trace.	Every result exposes its path rather than hiding it behind an answer.
Serving	LITE laptop mode and optional MAIN Kaggle/T4 mode through cloudflared.	Same backend in two places; the UI tries MAIN first then LITE.
Exports	Downloadable HTML report and JSON trace.	The HTML is printed to PDF from the browser; it is not a direct-PDF button.

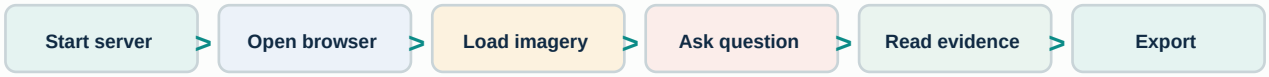
Important presentation truth

Do not overclaim
The repository does not contain the planned BigEarthNet fine-tune, RemoteCLIP training, or benchmark evaluation pipeline. Present it as a zero-shot Qwen3.5:2b VLM grounded by rule-based image tools, not as an already fine-tuned remote-sensing foundation model.

This is a web app, not a browser extension. Normal use is: start the server, open a browser, upload imagery, ask a question, then inspect and export evidence.

2. Fastest way to use the app

For a presentation, use local LITE mode first. It is the most reliable route because the React UI, FastAPI backend, sample files, and local Ollama model are part of the project environment.



One-time local setup

```
.venv\Scripts\python -m pip install -r requirements.txt
cd web
npm install
npm run build
cd ..
ollama pull qwen3.5:2b
```

The project notes recommend Python 3.12 in the existing .venv. The React build only needs to be rebuilt when the web UI changes. Ordinary use does not require editing source files.

Launch and open

```
ollama serve
.venv\Scripts\python -m uvicorn satquery.server:app --host 127.0.0.1 --port 8000
Open: http://127.0.0.1:8000
```

> **Before a live demo**
> Look at the top-right status. It should say LITE and show qwen3.5:2b online. If the VLM is offline, pixel measurements still run, but the written answer becomes an evidence-only fallback.

Six user actions

Step	What to do	What the app does
1	Put the first file in Image A.	Treats it as an optical scene or an earlier date unless you choose modality.
2	Optionally put a second file in Image B.	Interprets it as SAR or a later date after validation.
3	Choose Auto, Optical, or SAR.	Auto usually works; explicitly select for panchromatic or ambiguous inputs.
4	Write a concise question or choose a suggestion.	A keyword router chooses caption, grounding, change, or VQA behavior.
5	Click Run analysis or press Ctrl+Enter.	Validates, plans, measures, calls VLM if online, then renders evidence.
6	Read answer with image, numbers, and trace.	Lets you evaluate support before exporting.

3. Inputs: what to upload and how to label it

The app accepts one or two raster images. A pair only proceeds when its geometry is compatible. The validator refuses to guess when a pair is not co-registered.

Supported inputs and automatic interpretation

Input shape	Use it for	How the validator interprets it
One image	VQA, caption, grounding, water question.	single. Auto treats 1-2 bands as SAR and 3 or more as optical.
Two optical images	Before/after change questions.	bitemporal if size, CRS, overlap, and band count match. A is earlier; B is later.
Optical plus SAR	Joint water and built-up analysis.	optical_sar. The pair must have matching size and compatible coordinates.
Primary files	GeoTIFF / TIFF.	The intended format for real use; read in memory through Rasterio.
Benchmark-only files	PNG / JPG / JPEG.	Accepted for public benchmarks but without spatial CRS metadata.

Validation limits that matter

Check	Current rule	Why it matters
File count	Exactly 1 or 2.	The interface is designed around one scene or a pair.
Dimensions	Minimum side 8 px; maximum 300 million pixels per band.	Malformed or oversized raster headers are refused.
Processing size	Largest side downsampled to 1024 px.	Keeps laptop analysis responsive; full-resolution tiling is future work.
Band limit	At most 16 bands.	Covers common Sentinel-2 layouts while protecting memory.
Pair geometry	Same size; matching CRS and overlapping bounds when CRS exists.	Prevents meaningless change/fusion analysis.

Helpful band conventions
For 4-band Cartosat-2S MX imagery, the code maps B, G, R, NIR in that order. Twelve- and thirteen-band Sentinel-2 layouts also have built-in role maps. SAR is converted to dB and rendered as a false-color preview.

4. Ask the right question

The controller uses a transparent keyword classifier. Change, before, after, and between prefer change analysis; highlight, locate, and where prefer grounding; describe and caption prefer captioning. Other questions default to VQA.

Reliable presentation prompts

Scenario	Recommended question	Expected plan
Describe one scene	Describe the land-cover and major objects visible in this image.	vlm_caption; water tool is added if water is named.
Find water	Highlight the water body referred to in the query.	water_optical or water_sar, then vlm_ground.
Flood / change	What changed between these two dates, and where did the change occur?	change_cva, then vlm_change.
Optical plus SAR	Use the optical and SAR images together to identify built-up and water-covered regions.	water tools, water fusion, built-up fusion, then vlm_vqa.
Question and answer	Is there a river in this image?	Water tool for water questions, then vlm_vqa.

Advanced parameters

The Tool parameters box accepts JSON in the form `{"tool_name": {"parameter": value}}`. Only declared parameters of registered tools are used; unknown fields are ignored.

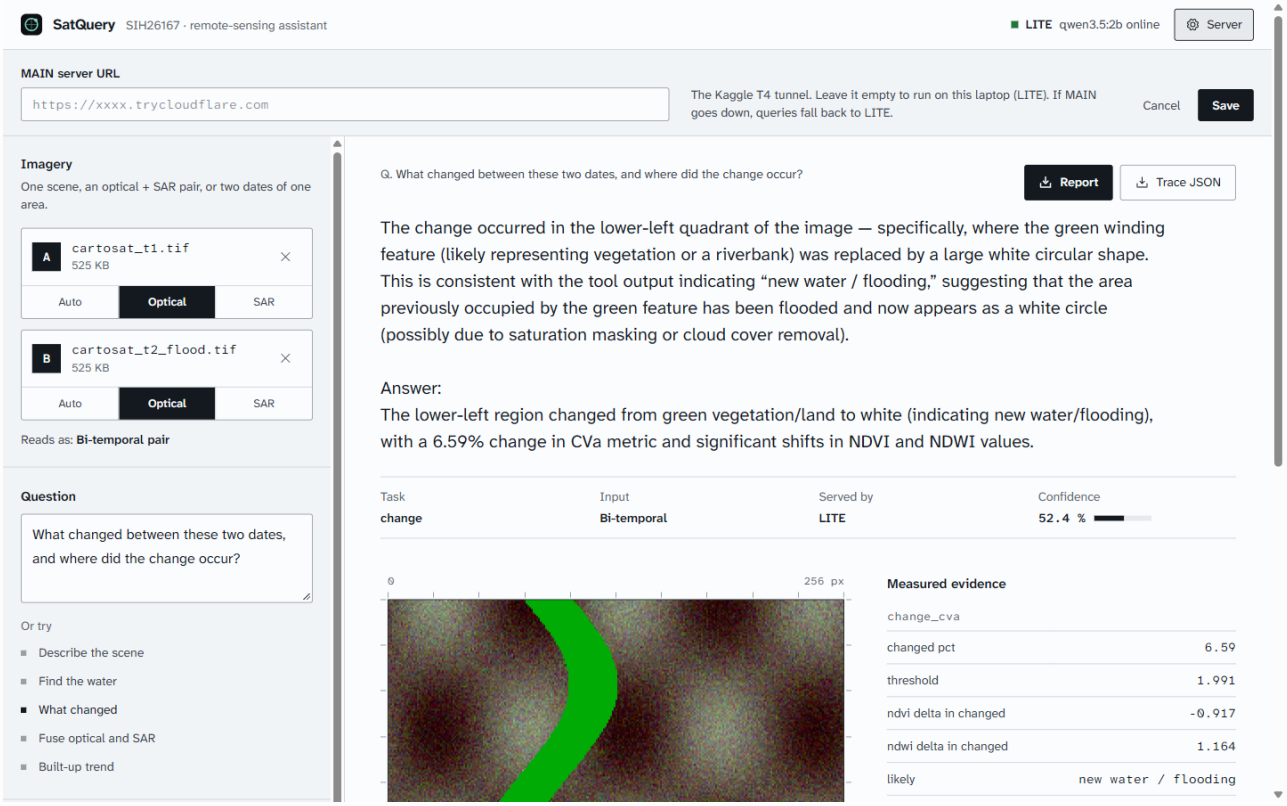
```
{"water_sar": {"threshold_db": -15}}
```

Use manual thresholds only when you understand source calibration. The default SAR water method uses Otsu thresholding, which adapts to a scene but can create a dark-class mask even where there is no true water.

> **Best practice**
> For a jury demo, use the supplied sample prompt language. Explain which evidence route the question activates rather than typing a long unpredictable prompt.

5. How to read the result screen

The result screen is built for inspection, not just a chatbot answer. The screenshot below is an existing local sample result. Treat prose as a model suggestion; use the measured evidence and mask layers as the defensible support.



Example only: two synthetic Cartosat-style dates in LITE mode. The lower table and trace are the evidence to point at during a demo.

Screen area	What it tells you	How to explain it
Answer	VLM-generated natural-language response.	Useful summary, but not proof by itself.
Task / Input / Served / Confidence	Selected task, data layout, MAIN or LITE, and aggregate confidence.	Show which route ran and whether VLM was available.
Image viewer	Original inputs plus evidence layers; masks are solid red.	Click layers. For change, show both dates and their evidence overlay.
Measured evidence	Tool statistics such as water percent, change percent, deltas, thresholds, agreement.	Use this to ground your claim.
Execution trace	Tool, status, time, params, stats, confidence, and output.	This is the auditable agent trace.

6. Outputs, reports, and the API

A completed result is kept in the browser response. The current code has no database, persisted job queue, or server-side report store.

What is produced

Output	Contents	Use
On-screen answer	Written response to the question.	Narrative only; pair it with measurements.
Evidence layers	PNG previews for inputs and each tool mask.	Switch layers; red pixels are detected/change evidence.
Evidence table	Statistics from successful tools.	Use exact values in a technical presentation.
Trace	Ordered tools, params, status, duration, stats, confidence, text.	Demonstrates reproducible orchestration.
satquery_report.html	Answer, evidence, inputs, trace table, visual evidence.	Human-readable report.
satquery_trace.json	Query, answer, confidence, evidence, trace.	Machine-readable audit artifact.

Turn the built-in report into a PDF

1	Click **Report**. The browser downloads satquery_report.html.
2	Open it in Chrome, Edge, or another browser.
3	Use Ctrl+P, select **Save as PDF**, then name it after the scene or case.
4	Keep the paired satquery_trace.json when traceability matters.

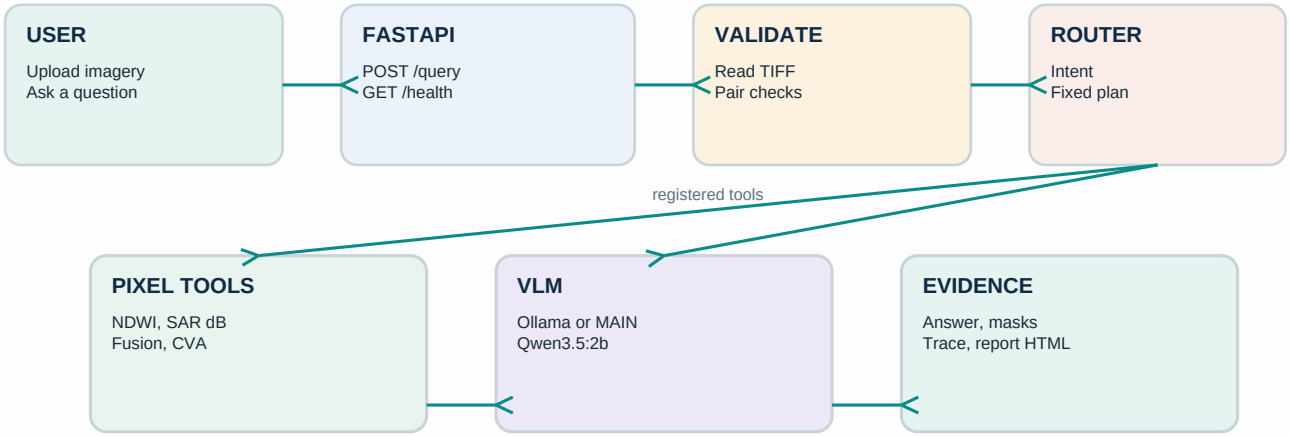
Important distinction
The Report button downloads HTML, not a native PDF. Browser printing is the intended way to create a PDF from the app output.

HTTP contract

Endpoint	Input / output
GET /health	Returns server status, VLM URL, model name, and whether the VLM endpoint responds.
POST /query	Multipart q, repeated files, optional comma-separated modalities, optional params JSON. Returns answer, evidence, confidence, trace, encoded images, and report_html.
GET /samples/*	Static synthetic demo files for the local primary UI, when mounted.

7. Technical architecture

The application stays deliberately compact: React browser client -> FastAPI -> raster validation -> fixed router and registry -> image tools plus VLM -> answer, masks, metrics, trace, and report HTML.



The plan is fixed and logged: an auditable controller, not an opaque multi-agent chain.

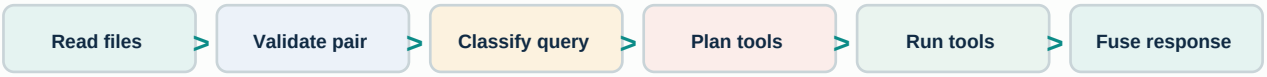
Technology stack

Layer	Technology	Role
Web client	React 19, Vite 8, Tailwind v4, Phosphor icons, Atkinson Hyperlegible.	Uploads, settings, results, trace viewer, downloads.
API service	Python, FastAPI, Uvicorn, python-multipart, httpx.	Endpoints, guards, health, report HTML, static UI.
Raster processing	Rasterio, NumPy, Pillow.	GeoTIFF reading, size rules, previews, masks, encoding.
Language model	Ollama-compatible OpenAI-style endpoint; default qwen3.5:2b.	Captioning, VQA, grounding boxes, change language.
Fallback UI	Gradio.	Alternative local UI; React is primary.
MAIN hosting	Kaggle T4 notebook plus cloudflared.	Optional remote GPU/VLM serving.

No hidden platform dependency
There is no LangChain, CrewAI, hosted database, vector database, or paid API required by the executable code. The agent behavior comes from a Python router and named tool registry.

8. What happens inside one request

The control flow is fixed and inspectable. A simple classifier selects a predefined plan. Every tool invocation writes parameters, status, duration, statistics, confidence, and text into the response trace.



Data flow

Stage	What happens	Why it matters
Read and normalize	Raster data is decoded in memory. Optical gets band roles; SAR is converted to dB. Display/VLM previews are stretched.	Tools and VLM see the same scene.
Validate configuration	Labels input as single, optical_sar, or bitemporal. Invalid geometry raises a user-facing error.	Avoids comparisons of unrelated images.
Classify and plan	Keywords choose intent, then configuration chooses a registered ordered tool list.	Simple enough to explain and audit.
Execute	Pixel and VLM tools run. Offline/error state remains visible.	Failure does not become a silent hallucination.
Render	Masks are painted red on previews. JSON includes answer, evidence, confidence, trace, images, and report HTML.	The UI can show provenance.

Specialist tools

Tool	What it does	When
water_optical	NDWI water mask.	Optical water query; fusion.
water_sar	Dark SAR dB / Otsu water mask.	SAR water query; fusion.
fuse_water	Intersection and agreement of both water masks.	Optical plus SAR.
builtup_fusion	Bright SAR VV constrained by low optical NDVI.	Optical plus SAR.
change_cva	Z-scored change vector, mask, NDVI/NDWI deltas, likely label.	Bi-temporal.
vlm_vqa / caption / ground / change	Text answer, caption, normalized box, or change description.	Task-specific VLM steps.

9. LITE, MAIN, security, and resilience

LITE runs on the laptop. MAIN runs the same backend in a Kaggle T4 notebook and exposes it with a temporary Cloudflare quick tunnel. The UI can remember MAIN URL/token in browser local storage and automatically fall back to LITE when MAIN is unreachable.

Choose the right mode

Mode	Where it runs	Use it for	Caution
LITE	Laptop FastAPI + local Ollama + CPU tools.	Reliable live demo or local testing.	Ollama must run and the model must be available.
MAIN	Kaggle T4 with Ollama + FastAPI + cloudflared.	Remote GPU capacity / later heavier model.	Tunnel is temporary and public; use its generated token.

Using MAIN

1	Open kaggle/main_server.ipynb in Kaggle and enable T4 GPU plus Internet.
2	Provide project code through the private Kaggle Dataset zip or a repository URL as the notebook documents.
3	Run All. It starts Ollama, model, Uvicorn, and cloudflared, then prints MAIN URL and TOKEN.
4	Click Server in the local UI, paste URL/token, save, and verify MAIN plus model online.
5	Stop the Kaggle session after the demo. The quick-tunnel URL changes every run.

Implemented safeguards

Control	Current behavior
Request limits	512 MB default POST body, 2,000-character query cap, and four concurrent queries by default.
Input parsing	Pinned GeoTIFF driver and safe PNG/JPEG handling protect against unsafe format interpretation.
Browser controls	CSP, nosniff, no-referrer, frame denial, trusted hosts, restricted CORS.
Remote access	SATQUERY_TOKEN protects POST /query when configured; notebook generates a fresh token.
VLM failure	The pipeline returns measured evidence and records vlm_offline instead of crashing.

Privacy
LITE keeps imagery local to the laptop stack. MAIN uploads files to the configured Kaggle tunnel. Do not share URL/token publicly or send sensitive imagery to unapproved external services.

10. Capability status and honest limits

A credible presentation distinguishes what the prototype demonstrates now from the roadmap. That is especially important because the project research document includes planned components that are not in the executable source.

Status against the SIH feature set

Requirement	Status	What to say
M1 - RS adaptation / fine-tune	Not implemented.	Planned: BigEarthNet QLoRA and RemoteCLIP safety net. Do not say it is trained.
M2 - single-image VQA	Implemented.	VLM answer with optional water evidence.
M3 - caption / grounding	Implemented.	Both exist; grounding is VLM box plus index evidence, not SAM refinement.
M4 - change VQA	Implemented.	CVA mask plus NDVI/NDWI deltas and VLM wording.
M5 - optical plus SAR	Implemented.	Water fusion, agreement, built-up heuristic, VLM answer.
M6 - agent trace	Implemented.	Fixed router/registry with per-step audit trace.
M7 - GUI/report	Implemented.	React UI, validation, evidence, confidence, HTML/JSON export.

Limits to state upfront

Limit	Practical implication
Model quality	The zero-shot VLM can hallucinate scene details or misstate numeric relationships. Measurements should anchor claims.
Validation data	Included demo GeoTIFFs are synthetic. Real Cartosat/RISAT-like data and VRSBench/RSVQA/CDVQA evaluation remain needed.
Spatial scale	Large scenes are reduced to 1024 px. Full-resolution tile-and-stitch is future work.
SAR calibration	Otsu always makes a dark class; manual thresholds need calibrated data. Built-up dB thresholds also need calibration.
Routing	Keyword intent and fixed pair plans are explainable but not broad natural-language planning.
Persistence	No user auth, database, or persistent report store.

Current-source note
Older handoff notes record two behavior failures. Current source includes a flat-change guard and renders change masks on both dates. Re-run behavior tests before the event for fresh evidence, but do not present the older 7/9 note as the current code state.

11. A strong 3-minute presentation

Use this sequence: problem -> proof -> traceability -> roadmap. Keep the audience on imagery and measurements, not a long model monologue.

Time	Show	Say
0:00-0:25	Title and a satellite scene.	Remote-sensing data is rich but hard to query quickly. SatQuery enables plain-language, evidence-backed questions.
0:25-0:55	Load Flood, two dates.	The system recognizes aligned optical dates and validates comparability before analysis.
0:55-1:30	Ask what changed and where.	It selects change-vector analysis first, then lets the VLM write a concise explanation with measured context.
1:30-2:00	Show red mask, measurements, trace.	The difference is evidence: changed pixels, NDVI/NDWI direction, threshold, timing, and tools.
2:00-2:25	Load Optical and SAR.	A second route fuses optical water evidence with SAR backscatter and measures agreement.
2:25-2:45	Click Report and Trace JSON.	Each result exports a human-readable report and machine-readable trace.
2:45-3:00	Closing roadmap.	Next: domain adaptation, formal evaluation, real Indian scenes, full-resolution processing.

Three lines worth memorizing

Opening
SatQuery is an evidence-first remote-sensing assistant: ask a question about optical, SAR, or change imagery and inspect the visual and numerical support behind the answer.

Differentiator
We do not ask a model to guess alone. A transparent router combines specialized image tools with a compact VLM and gives the jury the trace.

Honest close
The current prototype proves the workflow; the next milestone is domain adaptation and real benchmark validation.

Judge Q and A

Question	Accurate answer
Why agentic?	It selects a registered ordered plan, runs specialist tools, passes measurements onward, and records every call.
Why not just VLM?	NDWI, SAR dB analysis, fusion, and CVA yield inspectable pixel evidence.
What if model is offline?	The tools run and the trace marks VLM offline; it does not pretend to have a text answer.
Is confidence calibrated?	No. It is an aggregate of returned tool/model confidences, not a calibrated probability of correctness.
Is it fine-tuned?	Not yet. Zero-shot Qwen plus deterministic tools is what runs now; adaptation is next.

12. Handover checklist and project map

Pre-demo checklist

Check	Done when
Model health	Top bar says LITE or MAIN and model online.
Samples	Flood pair and optical-plus-SAR samples load, or files are already selected.
Network plan	LITE was tested; MAIN URL/token are ready only if you need remote mode.
Evidence	You know where the red overlay, measured evidence, and trace are.
Exports	One saved HTML/PDF report and trace JSON are on the presentation laptop.
Claims	The team knows not to claim M1 fine-tuning or real-data evaluation is completed.

Where responsibilities live

Path	Purpose
satquery/server.py	FastAPI, upload/security guards, health/query endpoints, static UI/sample mounting, report HTML.
satquery/validate.py	Formats, dimensions, modality, bands, SAR dB conversion, preview, pair checks.
satquery/router.py	Intent keywords, plan, tool execution, trace, evidence layers, result object.
satquery/tools.py	Water, fusion, built-up, change, VLM tools, registry.
satquery/vlm.py	OpenAI-compatible VLM request and confidence extraction.
web/src/App.jsx	Primary React uploads, result UI, evidence, trace, downloads, MAIN settings.
web/src/api.js	MAIN-first/LITE fallback and localStorage settings.
kaggle/main_server.ipynb	Kaggle T4 plus cloudflared server recipe.
samples/	Synthetic Cartosat-style and RISAT-style demo inputs.
test_smoke.py / test_behaviour.py	Synthetic end-to-end and behavior checks.

Final handover
Keep this guide, one exported report PDF, its trace JSON, and the demo GeoTIFF files together. During the event, show both the answer and the evidence every time.